# EARLY DETECTION OF LITHIUM DENDRITE RISK IN EV BATTERIES USING QUANTUM SUPPORT VECTOR MACHINES

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**Vishak Senthilkumar** **Aravindhan S** **Manesh Ram**

# ABSTRACT

The overall number of electric vehicles (EVs) on the roads is increasing and the need for the safety and reliability of their batteries is becoming more important. Dendrite formation is one of the most critical issues facing an electric vehicle's battery and can cause not only safety hazards but also destruction of the battery itself. This project offers a unique procedure for spotting battery dendrite risk early, using a hybrid machine learning model that merges both classical and quantum computing methods. Quantum Support Vector Machines classify the health of the battery time-series data, which consists of voltage, current and temperature. The sensors that are placed on the battery gather continuous data for a certain period of time and this data is then divided into smaller segments using a sliding window technique of pre-processing. For each segment, various statistics including average, variance and rate of change are computed to identify potential features in the battery's behavior. Features that have been selected are subsequently provided to the hybrid model where the QSVM further pushes the feature transformation process making it easier to detect very slight abnormalities which are signs of dendrite growth. It is the objective of this model to generate real-time, very precise predictions of dendrite risk that will facilitate quick actions taken and thus better management of the overall battery. The fusion of quantum computing and traditional machine learning techniques is what marks a key milestone for the EVs battery health monitoring sector.

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| | |
|---|---|
| **QSVM** | Quantum Support Vector Machine |
| **QML** | Quantum Machine Learning |
| **SVM** | Support Vector Machine |
| **EV** | Electric Vehicle |
| **QAOA** | Quantum Approximate Optimization Algorithm |
| **NISQ** | Noisy Intermediate-Scale Quantum |
| **PCA** | Principal Component Analysis |
| **SOC** | State of Charge |
| **CE** | Coulombic Efficiency |
| **Re** | Electrolyte Resistance |
| **Rct** | Charge Transfer Resistance |
| **BMS** | Battery Management System |
| **NASA** | National Aeronautics and Space Administration |

# CHAPTER 1

# INTRODUCTION

EVs can contribute to sustainable transportation and lithium-ion batteries are the key to their operation. The charging process subjects the EV batteries to different electrical and thermal conditions, which when allowed to get out of check may lead to abnormal behavior. Thermal runaway is one of the most important safety issues since the phenomenon, which is defined as an abrupt rise in temperature, may cause battery deterioration or system malfunction. Such abnormal conditions must therefore be identified early in order to be able to ensure that operations are safe and also to guarantee reliability of the system.

Due to the growing implementation of sensor-based battery management systems, nowadays it is possible to collect a considerable amount of time-stamped battery charging data in real time. This data is usually continuous measurements, including voltage, current, temperature and state of charge, measured during the charge, which are recorded during the charging cycle. This data is usually analyzed with machine learning methods and used to identify abnormal charging conduct. Nevertheless, most of the current methods are based on classical machine learning models only, and these might not be able to depict the complicated nonlinear relationships that can exist in multivariate time-series sensor data.

The new developments in quantum computing provided quantum-inspired machine learning (QML) methods that present alternative data representation and transformation techniques. Despite the fact that modern quantum devices have not offered any computational advantage of practical datasets, hybrid learning systems with classical and quantum-inspired components have been considered, because the feature representation can be improved. Here, it is assumed that QML will not replace the classical machine learning, but it represents another processing layer that can be added to the already existing learning pipeline.

This paper suggests a hybrid classical-quantum learning model to detect anomalies in EV battery charging systems with real time stamped sensor data. It is based on the suggested procedure that combines the classical time-series preprocessing and feature extraction with a quantum-inspired processing layer that is aimed at executing nonlinear feature transformation. The resulting hybrid feature representation is then studied with a classical classifier to determine abnormal charging patterns that relate to thermal runaway.

This project will have the main goal of designing and deploying an entire learning pipeline showing the extent to which advanced machine learning techniques can be integrated in a safety-critical and real world application. Instead of prioritizing the superior performance or computational acceleration, it focuses on the architectural design, feasibility, and the systematization of evaluation of hybrid feature representations of time-series anomaly detection. The efficiency of the suggested framework is assessed with the help of the data set of time-stamped sensor readings that were recorded in EV battery charge and thermal runaway events.

# CHAPTER 2

# LITERATURE REVIEW

Electric vehicles rely heavily on lithium-ion batteries for energy storage. Ensuring the safety and reliability of these batteries is therefore a critical research problem. One of the major challenges in battery management is the early detection of abnormal behaviour such as lithium dendrite formation. Detecting these patterns requires analyzing large amounts of time-series sensor data, including voltage, current, temperature, and state-of-charge measurements.

In recent years, machine learning techniques have been widely used for battery health monitoring and anomaly detection. At the same time, advances in quantum computing have introduced new approaches for improving machine learning models through quantum feature mapping and quantum kernel methods. This chapter reviews the existing research related to quantum support vector machines, hybrid quantum-classical learning frameworks, and anomaly detection in battery systems.

## 2.1 QUANTUM KERNEL METHODS IN SUPPORT VECTOR MACHINES

Support Vector Machines are commonly used for classification tasks due to their ability to construct decision boundaries between different classes of data. Traditional SVM models rely on classical kernel functions such as polynomial or radial basis kernels to map data into higher dimensional feature spaces.

Recent work by Abdulsalam and Ahmad explored the use of quantum kernels within the SVM framework [1]. Their study compared classical kernel functions with quantum kernel functions and analyzed their performance across several classification tasks. The results showed that quantum kernels can sometimes improve the separability of complex datasets by mapping data into richer feature spaces.

The findings from this work provide useful insights into how kernel functions influence classification performance. In the proposed project, this concept is adopted by using a quantum-inspired kernel within a QSVM model to improve the separability of battery degradation patterns extracted from sensor data.

## 2.2 QUANTUM FEATURE MAPPING FOR CLASSIFICATION

An important component of quantum machine learning models is the process of encoding classical data into quantum states. Different quantum feature maps can significantly influence how well a QSVM classifier performs.

Pramanik and Jha evaluated multiple quantum feature maps for classification tasks using QSVM [4]. Their research investigated Pauli-based feature maps and analyzed their effectiveness for binary classification problems. The study demonstrated that selecting an appropriate feature encoding method plays an important role in improving model accuracy.

This work motivates the use of quantum-inspired feature mapping in the proposed system. By encoding classical battery features into a higher dimensional quantum feature space, the QSVM classifier can better capture nonlinear relationships within the sensor data.

## 2.3 HYBRID CLASSICAL-QUANTUM LEARNING FRAMEWORKS

Several researchers have proposed hybrid architectures that combine classical machine learning techniques with quantum models. In such systems, classical algorithms perform preprocessing and feature extraction, while quantum models perform the final classification.

Yi proposed a hybrid quantum-classical framework for traffic state classification [6]. In this work, classical feature extraction techniques were applied to traffic sensor data, and the processed features were then used as inputs for a QSVM classifier. The results showed that combining classical preprocessing with quantum classification can improve performance for complex datasets.

Similarly, Roth et al. introduced the AutoQML framework, which automates the design of quantum machine learning pipelines [5]. The framework automatically selects quantum feature maps, tunes model parameters, and constructs hybrid learning pipelines. Although this approach simplifies the model development process, it also introduces additional computational overhead.

These studies demonstrate that hybrid quantum-classical architectures can provide flexible and powerful learning systems. The proposed work follows a similar hybrid approach by combining classical feature engineering with quantum-inspired feature mapping and QSVM classification.

## 2.4 QUANTUM MACHINE LEARNING FOR BATTERY MONITORING

Machine learning methods have also been explored for battery monitoring and anomaly detection applications. In particular, researchers have begun investigating quantum machine learning techniques for analyzing battery sensor data.

A study by P., V., and G. proposed a real-time quantum machine learning approach for detecting defective cells in lithium-ion battery packs [3]. Their system used quantum-enhanced clustering techniques to identify abnormal battery behaviour using voltage, current, and temperature measurements. Although their approach demonstrated promising results, it relied primarily on clustering-based anomaly detection rather than supervised classification models.

Another related work by Dutta et al. explored the use of quantum kernel based support vector machines with Quantum Approximate Optimization Algorithm embedding for medical data classification [2]. Their results showed that quantum feature embedding can improve classification performance when compared with traditional SVM models.

The proposed project takes inspiration from these works by applying QSVM classification to battery degradation analysis. Instead of clustering methods, the system uses supervised classification to predict battery risk levels based on engineered features derived from battery sensor data.

## 2.5 RESEARCH GAP

Although existing research demonstrates the potential of quantum machine learning models, several limitations still remain. Many studies rely on simulated quantum environments rather than real quantum hardware, which limits their practical evaluation. Additionally, several approaches focus on clustering-based anomaly detection instead of supervised classification techniques that can provide clearer decision boundaries.

Another limitation is that many quantum machine learning studies evaluate their models on relatively small or domain-specific datasets.

Real-world battery monitoring systems generate large amounts of multivariate time-series data, which require specialized preprocessing and feature engineering techniques.

To address these challenges, the proposed work develops a hybrid classical-quantum framework for early detection of lithium dendrite risk in EV batteries. The system integrates classical feature extraction, quantum-inspired feature mapping, and QSVM classification to analyze battery degradation patterns and predict risk levels.

# CHAPTER 3

# SYSTEM DESIGN

## 3.1 DESIGN OVERVIEW

The proposed system design aims at developing a dependable hybrid Quantum Support Vector Machine (QSVM)-based system for detecting anomalies and assessing risks in electric vehicle (EV) battery charging systems. The architecture is inspired by existing quantum machine learning-based battery anomaly detection solutions, particularly those based on unsupervised clustering algorithms such as K-means.

Although clustering-based methods can identify coarse deviations in battery behaviour, they often fail to detect subtle early-stage anomalies and lack clear decision margins. To overcome these limitations, the proposed system replaces clustering with a margin-based hybrid QSVM classifier combined with a risk-oriented decision layer.

The system follows a modular pipeline architecture in which each module performs a specific transformation and passes its output to the next stage. This design improves interpretability, scalability, and ease of integration with existing battery management systems.

## 3.2 SYSTEM OVERVIEW

The proposed EV battery monitoring framework follows a sequential processing pipeline that transforms raw sensor measurements into battery risk predictions.

The overall workflow of the system is illustrated in Figure 3.1.

The process begins with acquiring battery sensor measurements and proceeds through preprocessing, feature engineering, quantum feature transformation, and QSVM-based classification.

The major stages of the workflow are summarized as follows:

1. **Data Acquisition:** Battery sensor data such as voltage, current, temperature, and state of charge are collected during charging and discharging cycles.
2. **Data Preprocessing:** Raw measurements are filtered to remove noise, normalized to maintain consistent scales, and segmented using sliding windows to capture short-term battery behaviour.
3. **Feature Engineering:** Statistical and electrochemical features are extracted from each window to represent battery behaviour in a structured feature space.
4. **Quantum Feature Mapping:** Classical feature vectors are mapped into a higher-dimensional Hilbert space using quantum-inspired encoding techniques.
5. **QSVM Classification:** A hybrid Quantum Support Vector Machine classifier learns decision boundaries between different battery risk levels.
6. **Risk Assessment:** The predicted QSVM output is interpreted and mapped to battery risk categories, producing monitoring reports and confidence scores.

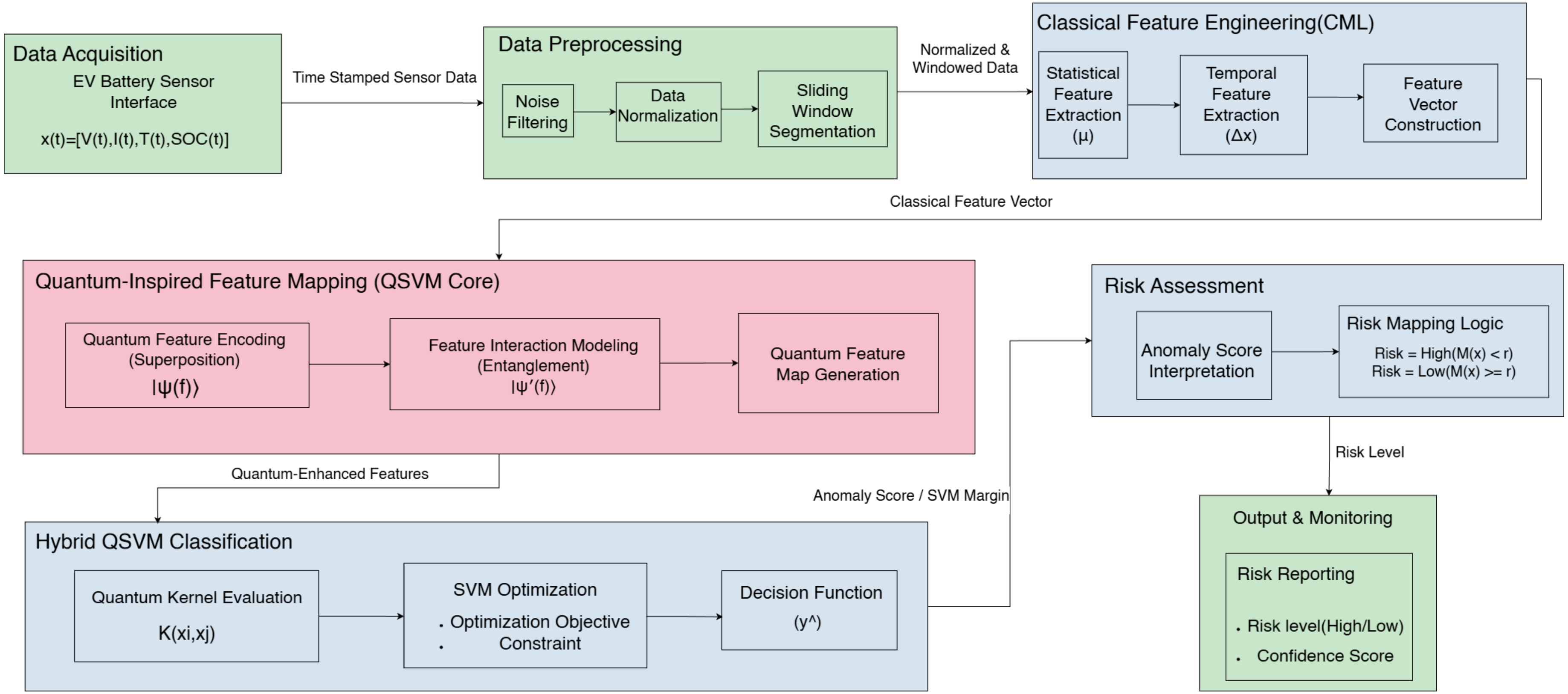


**Figure 3.1** Architecture of the Proposed QSVM-Based EV Battery System

### 3.2.1 Data Acquisition Module

The data acquisition module gathers time-stamped sensor data from EV battery packs during charging cycles. Each record contains synchronized measurements of:

- Voltage
- Current
- Temperature
- State of Charge (SOC)

These parameters collectively form a multivariate time-series representation of battery behaviour. Since battery anomalies often result from interactions between multiple parameters rather than single threshold violations, collecting multiple sensor signals provides a more comprehensive monitoring framework.

**Algorithm 1** Battery Data Loading Module

**Input:** NASA battery .mat files
**Output:** Structured dataset $D$
**for** each battery file $B_i$ **do**
  Load $B_i$ using MATLAB reader
  Extract cycle structure
  Separate discharge, charge, impedance cycles
  Store structured data in $D$
**end for**
**return** $D$

### 3.2.2 Data Preprocessing Module

Raw sensor data may contain noise, outliers, and variations in measurement scales. Therefore, preprocessing is necessary before feature extraction.

The preprocessing module performs:

- Noise filtering

- Data normalization
- Sliding window segmentation

Noise filtering reduces sensor disturbances, normalization ensures comparable feature ranges, and sliding window segmentation divides the time-series data into fixed-length windows that represent short-term charging behaviour.

**Algorithm 2** Sliding Window Degradation Modeling

```
Input: Feature matrix F
Output: Windowed dataset W
Define window size w and stride s
for each battery b do
    for each window segment do
        Compute capacity trend (slope)
        Compute capacity change (last - first)
        Compute impedance trend
        Compute degradation variance
        Store window features
    end for
end for
return W
```

### 3.2.3 Classical Feature Engineering Module

After preprocessing, statistical feature engineering is applied to extract meaningful descriptors from each window.

Typical features include:

- Mean
- Variance
- Rate of change
- Stability indicators

These features form a structured classical feature vector representing the battery behaviour within a time window. This stage reduces the

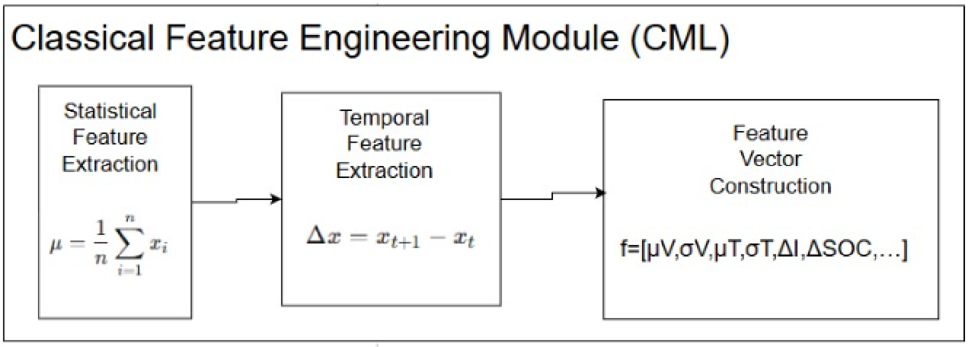


**Figure 3.2** Classical Feature Engineering Module

dimensionality of raw sensor data and provides interpretable inputs for further analysis.

---

**Algorithm 3** Statistical and Electrochemical Feature Extraction

---

**Input:** Dataset $D$
**Output:** Feature matrix $F$
**for** each discharge cycle $c$ in $D$ **do**
  Extract capacity
  Compute voltage statistics (mean, std, min, max, range)
  Compute current statistics (mean, std)
  Compute temperature statistics (mean, std, min, max)
  Extract impedance parameters (Re, Rct)
**end for**
Compute Coulombic Efficiency:
$CE = \frac{Q_{discharge}}{Q_{charge}}$
Merge all features into $F$
**return** $F$

---

As illustrated in Figure 3.2, the classical feature engineering stage extracts statistical and temporal features from battery sensor data.

### 3.2.4 Quantum-Inspired Feature Mapping Module

The quantum-inspired feature mapping module converts classical feature vectors into a higher-dimensional Hilbert space using quantum encoding techniques.

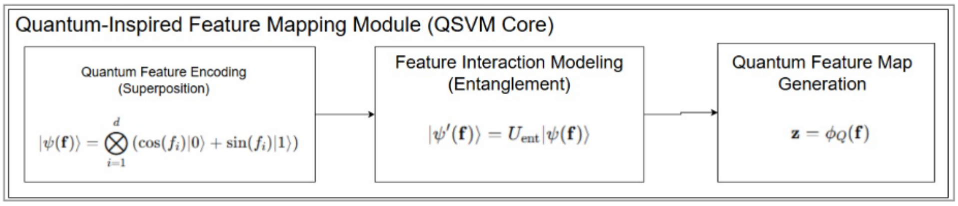


**Figure 3.3** Quantum-Inspired Feature Mapping Module

Initially, classical features are encoded using superposition-based representations, allowing multiple components to be represented simultaneously. Entanglement-inspired transformations are then applied to capture interactions between features.

These transformations increase the expressive capability of the feature space without requiring physical quantum hardware.

### 3.2.5 Hybrid QSVM Classification Module

The proposed system uses a hybrid QSVM classifier to detect anomalous charging behaviour.

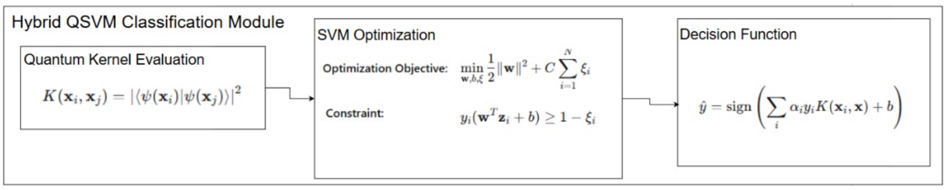


**Figure 3.4** Hybrid Quantum Support Vector Machine Classification Module

Unlike clustering-based approaches, QSVM learns a margin-based decision boundary that separates different battery degradation patterns in the transformed feature space.

The similarity between feature vectors is computed using a quantum kernel:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

The QSVM combines quantum kernel feature mapping with classical SVM optimization.

### 3.2.6 Risk Assessment and Output Generation

The system includes a risk assessment module that interprets the QSVM output to classify battery charging behaviour into different risk levels.

The classification output generated by the QSVM model is mapped to predefined battery health categories based on degradation patterns.

The battery state is categorized into the following risk levels:

- Very Low Risk
- Low Risk
- Medium Risk
- High Risk
- Critical Risk

These categories represent increasing levels of battery degradation and potential dendrite formation.

The final system outputs include:

- Predicted risk level
- Confidence score
- Monitoring reports

## 3.3 DESIGN SUMMARY

The proposed system extends existing battery anomaly detection frameworks by incorporating quantum-inspired feature mapping within a classical machine learning pipeline. The use of a hybrid QSVM classifier enables better separation between normal and abnormal bat-

**Table 3.1** Battery Risk Level Categories

| Class Label | Risk Level |
|---|---|
| 0 | Very Low Risk |
| 1 | Low Risk |
| 2 | Medium Risk |
| 3 | High Risk |
| 4 | Critical Risk |

tery behaviour while maintaining interpretability and computational efficiency.

# CHAPTER 4

# Implementation and Results

## 4.1 OVERVIEW OF THE PROPOSED ARCHITECTURE

The proposed hybrid Quantum Machine Learning framework was implemented using a modular pipeline that combines classical signal processing with quantum-enhanced machine learning techniques.

The overall workflow of the system includes the following stages:

1. Environment Setup
2. Data Loading
3. Timestamp Alignment
4. Missing Value Handling
5. Statistical Feature Engineering
6. Dendrite-Oriented Feature Engineering
7. Sliding Window Modelling
8. Window-Level Feature Construction
9. Quantum Feature Mapping
10. Quantum Support Vector Machine Classification
11. Risk Assessment and Reporting

Each module performs a specific transformation on the battery sensor data before passing it to the next stage of the pipeline.

## 4.2 IMPLEMENTATION ENVIRONMENT

The implementation was carried out using the Python programming language with several scientific computing and quantum computing libraries.

**Table 4.1** Software Environment Used for Implementation

| Tool / Library | Purpose |
|---|---|
| Python | Core programming language |
| NumPy | Numerical computation |
| Pandas | Data processing |
| Scikit-learn | Classical machine learning utilities |
| Matplotlib / Seaborn | Data visualization |
| Qiskit | Quantum circuit simulation |
| Qiskit Machine Learning | Quantum kernel and QSVM implementation |
| Google Colab | Development and execution environment |

The implementation was executed using the Google Colab platform, which allows integration of Python libraries and cloud-based computational resources.

## 4.3 DATASET DESCRIPTION

The NASA Lithium-Ion Battery Dataset was used for this study. The dataset consists of battery cycle measurements collected during laboratory experiments.

Each battery file contains multiple operational cycles including:

- Charge cycles
- Discharge cycles
- Impedance measurements

Battery measurements include the following parameters:

- Voltage
- Current
- Temperature
- Capacity
- Time

**Table 4.2** Dataset Summary

| Parameter | Value |
|---|---|
| Total Battery Files | 35 |
| Valid Batteries Processed | 33 |
| Total Cycles Loaded | 7313 |
| Discharge Cycles Used | 4600 |
| Valid Cycles after Filtering | 2667 |

## 4.4 DATA LOADING

Battery data stored in MATLAB (.mat) format was loaded using the `scipy.io` library. The MATLAB structures were parsed to extract discharge cycle measurements.

Only discharge cycles were used because they contain the relevant electrochemical behavior needed to analyze battery degradation patterns.

## 4.5 TIMESTAMP ALIGNMENT

Battery cycles contain time arrays representing measurements recorded during each cycle. These timestamps were aligned with cycle indices to maintain consistency across multiple battery datasets.

```
Aligning timestamps across batteries...
Timestamp alignment complete: 7,313 cycles aligned
```

**Figure 4.1** Timestamp Alignment Across Battery Cycles

Timestamp alignment enables proper comparison of degradation patterns across batteries.

## 4.6 MISSING VALUE HANDLING

Different cycle types contain different measurement fields. Therefore, missing values were handled using safe extraction techniques.

```
••• Extracting raw data with missing value handling...
    Raw data extraction complete: 4,600 cycles

    Missing value summary:
      capacity: 2692 missing
      voltage: 0 missing
      current: 0 missing
      temperature: 0 missing
      time: 0 missing
```

**Figure 4.2** Handling Missing Values in Battery Dataset

Median-based imputation was applied to fill missing entries while preserving statistical characteristics of the dataset.

## 4.7 STATISTICAL FEATURE ENGINEERING

Time-series battery signals were transformed into statistical descriptors to provide structured inputs for machine learning models.

The following statistical features were extracted:

- Mean
- Standard deviation
- Minimum value
- Maximum value
- Median

Electrochemical impedance parameters such as resistance components were also extracted to capture degradation behavior.

## 4.8 DENDRITE-ORIENTED FEATURE ENGINEERING

Additional electrochemical indicators were extracted to detect early signs of dendrite formation.

One important feature used is the differential capacity curve defined as

$$\frac{dQ}{dV} \tag{4.1}$$

Changes in the differential capacity curve can indicate structural degradation within the battery.

Additional descriptors extracted include:

- Skewness
- Kurtosis
- Entropy
- Interquartile Range

These features capture subtle variations in battery behavior that may indicate early dendrite growth.

## 4.9 SLIDING WINDOW MODELLING

Battery degradation is a gradual process. To capture temporal degradation patterns, sliding window segmentation was applied across battery cycles.

```
Window parameters calculated:
  Average degradation per cycle: -1.7310%
  Optimal window size: 10 cycles, stride: 3 cycles

Multi-scale configuration:
  Scale 1: Window=  5, Stride=  2, Overlap= 60.0%
  Scale 2: Window= 10, Stride=  3, Overlap= 70.0%
  Scale 3: Window= 20, Stride=  6, Overlap= 70.0%
Creating sliding windows for 33 batteries...
  Progress: 5/33 batteries
  Progress: 10/33 batteries
  Progress: 15/33 batteries
  Progress: 20/33 batteries
  Progress: 25/33 batteries
Sliding windows created: 352 windows
```

**Figure 4.3** Sliding Window Segmentation for Temporal Degradation Analysis

Within each window, degradation trends such as capacity slope and degradation variance were computed.

## 4.10 WINDOW-LEVEL FEATURE CONSTRUCTION

For each sliding window segment, several degradation trend features were extracted.

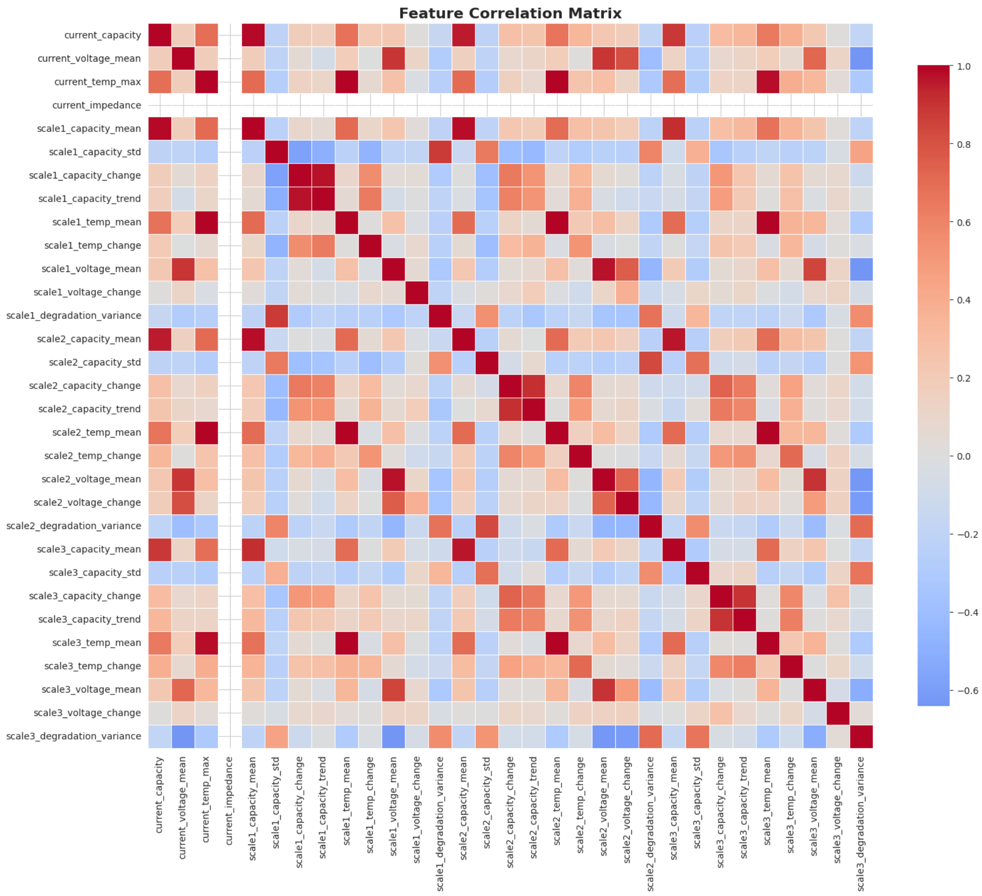


**Figure 4.4** Feature Extraction from Sliding Window Segments

These window-level features represent temporal degradation patterns observed in battery cycles.

## 4.11 QUANTUM FEATURE MAPPING

Principal Component Analysis (PCA) was used to reduce the dimensionality of the classical feature space before quantum encoding.

The original feature space was reduced to four principal components. These components were normalized to the interval $[0, \pi]$ for quantum circuit rotations.

Quantum feature encoding was then applied using parameterized quantum gates and entanglement operations.

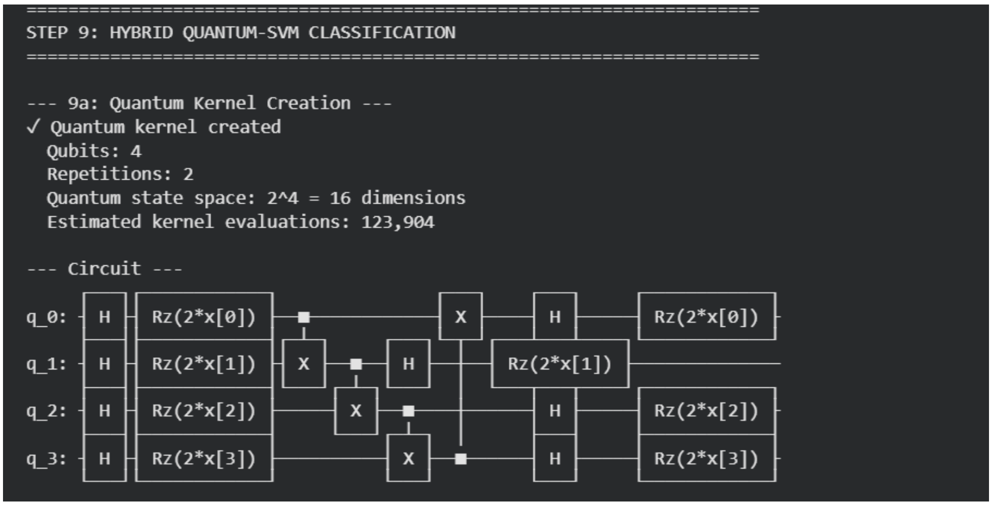


**Figure 4.5** Quantum Kernel Circuit Used for QSVM Feature Encoding

The quantum circuit used for kernel generation in the QSVM model is shown in Figure 4.5.

## 4.12 QUANTUM KERNEL SUPPORT VECTOR MACHINE

The classification stage uses a Quantum Kernel Support Vector Machine (QSVM).

The quantum feature map transforms classical feature vectors into quantum states as follows:

$$|\psi(x)\rangle = U_{\phi}(x)|0\rangle \tag{4.2}$$

The similarity between two samples is computed using the quantum kernel function:

$$K(x_i, x_j) = |\langle \psi(x_i)|\psi(x_j)\rangle|^2 \tag{4.3}$$

The resulting quantum kernel matrix is used by a classical SVM classifier to separate battery risk categories.

## 4.13 TRAIN-TEST SPLIT

The dataset was divided into training and testing sets to evaluate the model performance.

**Table 4.3** Dataset Split

| Dataset Portion | Percentage |
|---|---|
| Training Set | 80% |
| Testing Set | 20% |

## 4.14 TEST CASES

Several test scenarios were created to validate the ability of the model to detect different battery degradation states.

**Table 4.4** Test Case Examples

| Test Case | Input Condition | Expected Risk Level |
|---|---|---|
| TC1 | Stable voltage and temperature | Very Low Risk |
| TC2 | Slight voltage fluctuation | Low Risk |
| TC3 | Moderate degradation trend | Medium Risk |
| TC4 | Rapid capacity decline | High Risk |
| TC5 | Severe degradation indicators | Critical Risk |

## 4.15 RESULTS AND DISCUSSION

The hybrid quantum-classical framework successfully identified degradation patterns within the battery dataset. The quantum feature transformation improved the separability of battery health states in the feature space.

The predicted risk distribution and confidence levels obtained from the QSVM model are shown in Figure 4.6.

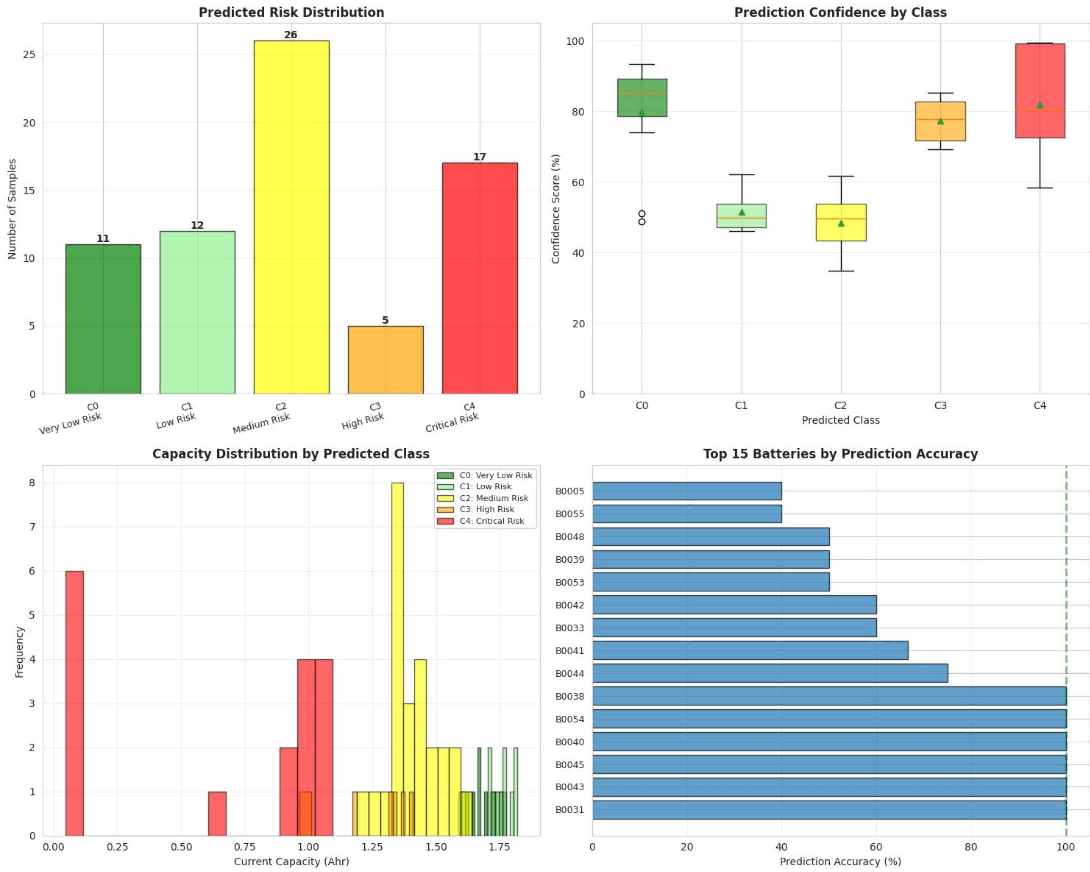


**Figure 4.6** Predicted Risk Distribution, Prediction Confidence and Battery Accuracy Analysis

## 4.16 PERFORMANCE METRICS

To evaluate the effectiveness of the proposed system, several performance metrics were used.

### 4.16.1 Accuracy

Accuracy measures the proportion of correctly classified samples among all test instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.4}$$

where $TP$, $TN$, $FP$, and $FN$ denote true positives, true negatives, false positives, and false negatives respectively.

### 4.16.2 Recall

Recall evaluates the model's ability to correctly identify actual anomalous battery conditions.

$$Recall = \frac{TP}{TP + FN} \tag{4.5}$$

High recall is critical in EV battery monitoring because missing an anomaly may lead to severe safety risks.

### 4.16.3 Feature Space Separability

Feature Space Separability (FSS) measures the separation between normal and anomalous samples after quantum feature transformation.

$$FSS = \frac{\|\mu_{normal} - \mu_{anomaly}\|_2}{\sigma_{normal} + \sigma_{anomaly}} \tag{4.6}$$

Higher values of FSS indicate better class separability.

### 4.16.4 Variance Amplification Ratio

Variance Amplification Ratio (VAR) quantifies the increase in feature variance produced by the quantum transformation.

$$VAR = \frac{Var(Q(x))}{Var(x)} \tag{4.7}$$

where $x$ represents the classical feature vector and $Q(x)$ represents the quantum-transformed feature vector.

Figure 4.7 presents the confusion matrix and per-class F1 score of the QSVM classifier.

The results demonstrate that the hybrid quantum-classical learning framework improves the representation of battery degradation patterns and enables early detection of lithium dendrite risk.

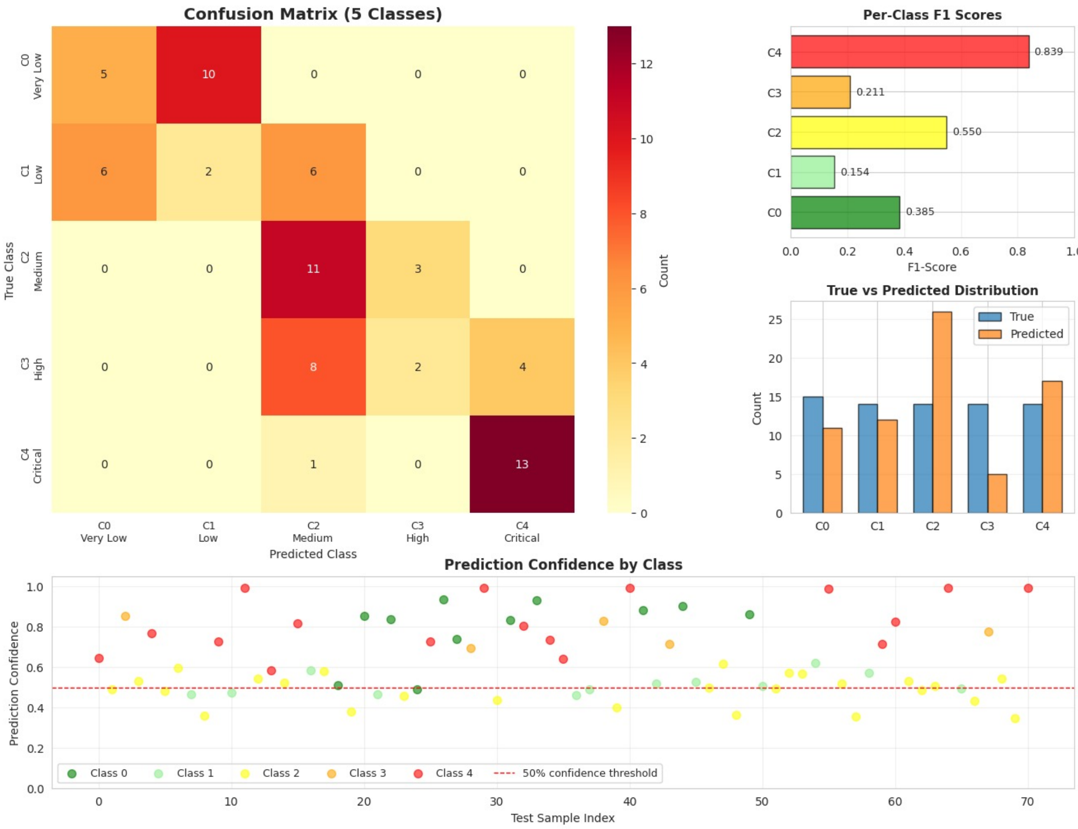


**Figure 4.7** Confusion Matrix and Per-Class F1 Score of the QSVM Model

# CHAPTER 5

# CONCLUSION AND FUTURE WORK

## 5.1 SUMMARY OF THE WORK

This powerful data processing pipeline established a cornerstone. We successfully converted raw, heterogeneous battery data into a standardized, clean, and feature-rich dataset. These multi-scale temporal features now prepare for the next phase of this project: the development and training of Quantum Machine Learning models for predictive analytics, with the main focus on early detection and understanding of battery dendrite formation to enhance battery safety, life, and performance. Feature vectors generated at this step are the required input for advanced machine learning algorithms and allow the study of subtle signatures of degradation that might indicate dendrite growth.

## 5.2 QUANTUM MACHINE LEARNING ENHANCEMENTS

Future improvements to the system can enhance quantum feature representation and classification performance.

### 5.2.1 Expand Qubit Count

Increasing the number of qubits from 4 to 6-8 would expand the feature space to 64-256 dimensions.

Maintaining PCA variance between 80-90% may improve kernel expressiveness and reduce support vector ratios.

### 5.2.2 Advanced Quantum Feature Maps

Future implementations may replace the existing feature map with advanced circuits such as:

- Pauli Feature Maps
- IQP circuits

Trainable quantum kernels can also be introduced to maximize class separability.

### 5.2.3 Hybrid Variational Quantum Circuits

Variational Quantum Eigensolver (VQE) or Quantum Neural Networks may be integrated before the SVM classifier to learn latent quantum representations.

## 5.3 CLASSICAL MACHINE LEARNING ENHANCEMENTS

### 5.3.1 Improved Label Design

Future work may incorporate physics-based thresholds such as State of Health (SOH) limits instead of statistical quantile labels.

### 5.3.2 Enhanced Feature Engineering

Additional electrochemical features such as Electrochemical Impedance Spectroscopy (EIS) parameters may be included.

Wavelet transforms can also be used to extract multi-frequency degradation signatures.

### 5.3.3 Classical Baseline Comparison

Benchmarking against classical models such as:

- Random Forest
- XGBoost
- LSTM

can help evaluate the practical advantages of quantum models.

## 5.4 DATA AND SCALABILITY

Future experiments may include larger datasets from NASA and CALCE battery repositories to improve generalization.

Testing on real quantum hardware using IBM Quantum devices can also evaluate practical quantum advantage.

## 5.5 SYSTEM INTEGRATION AND DEPLOYMENT

The risk prediction module can be integrated into Battery Management Systems (BMS) as a REST API microservice.

Real-time dashboards can visualize battery risk levels and trigger alerts when critical thresholds are reached.

Explainability methods such as SHAP can be used to interpret feature contributions in the model predictions.

## 5.6 MULTI-CHEMISTRY EXTENSION

Future systems may extend the methodology to other battery chemistries such as lithium-sulfur or solid-state batteries, enabling broader battery safety monitoring applications.

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